

Repeating Efficiently

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check the first test for Loops and stopping rules. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Loops and stopping rules. Proposed explanation: “Every loop runs at least once”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer to a countdown

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count iterations. Repeat from 6 and from 1. Explain why the final value need not equal 1. What happens from 7 if subtract 2 is replaced by subtract 0?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

6. Solve the given case

Repeat take the next unstamped card and stamp it four times.

How many cards are stamped? What assumption matters?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Repeat clap then tap three times. Write the complete sequence. How many actions occur? What would happen if the stopping instruction were missing?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 11 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use a loop to process each item card in the test pile, with an explicit stop when no cards remain.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Trace every test

Start counter at 2. While counter is below 8, add 2. When false, report and stop. Record the value and result at every test, including the final false test. State the number of iterations and the reported value.

My reasoning and evidence

13. W2 Check the first test

Use the same below 8 loop with add 2. Make separate runs from 8 and 9. Explain how many times the body runs and what each run reports. Why would adding 2 once be an error?

My reasoning and evidence

14. W3 Find the extra repetition

Start counter at 3. While counter is below 8, add 2. A learner writes values 3, 5, 7, 9, 11 and reports 11. Identify the first unauthorised update. Give the correct final test, reported value and iteration count.

My reasoning and evidence

15. W4 Transfer to a countdown

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count iterations. Repeat from 6 and from 1. Explain why the final value need not equal 1. What happens from 7 if subtract 2 is replaced by subtract 0?

My reasoning and evidence

16. W5 Test a proposed stopping rule

A procedure says start at 4; while counter is not equal to 9, add 2; report and stop. Write the first four values tested. Explain why the loop will not reach 9 using whole numbers. Suggest a changed condition that would stop once the counter reaches or passes 9, and trace that repaired run.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Repeating Efficiently

A fictional paper counter advances through a display until its condition becomes false.

Start at 2. While the current counter is below 10, add 3.

Test before each addition; report when the condition is false.

Condition checks before additions

	Current	Below 10?	Next
Check 1	2	true	5
Check 2	5	true	8
Check 3	8	true	11
Check 4	11	false	stop

Read a cell using both its row and column labels.

Repeating Efficiently

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First test

$2 < 10$ is true: update to 5.



2. Repeat

$5 < 10$: update to 8. $8 < 10$: update to 11.



3. Stop test

$11 < 10$ is false; do not add again.



4. Report

Final counter 11 after three additions.

CHANGE ONE THING

Start at 10 instead of 2.

Explain the changed result using the matching lesson.

Trace a fresh start of 9. How many additions occur, and must the report equal 10?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Give any repeated sorter step an explicit stopping condition and trace a zero-iteration example.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Give any repeated sorter step an explicit stopping condition and trace a zero-iteration example.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper counter advances through a display until its condition becomes false.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Final counter 11 after three additions.

1. Report
2. First test
3. Repeat
4. Stop test

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report Final counter 11 after three additions. The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure. Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Practice 2: explain the deciding evidence

Changed-case practice: Start at 10 instead of 2. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The loop runs zero times and reports 10.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The loop runs zero times and reports 10. Its first condition is already false, so no addition happens.

Repair a missing building block

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which input detects the difference between the two written conditions? Explain its entire run.. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early input.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention. Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
